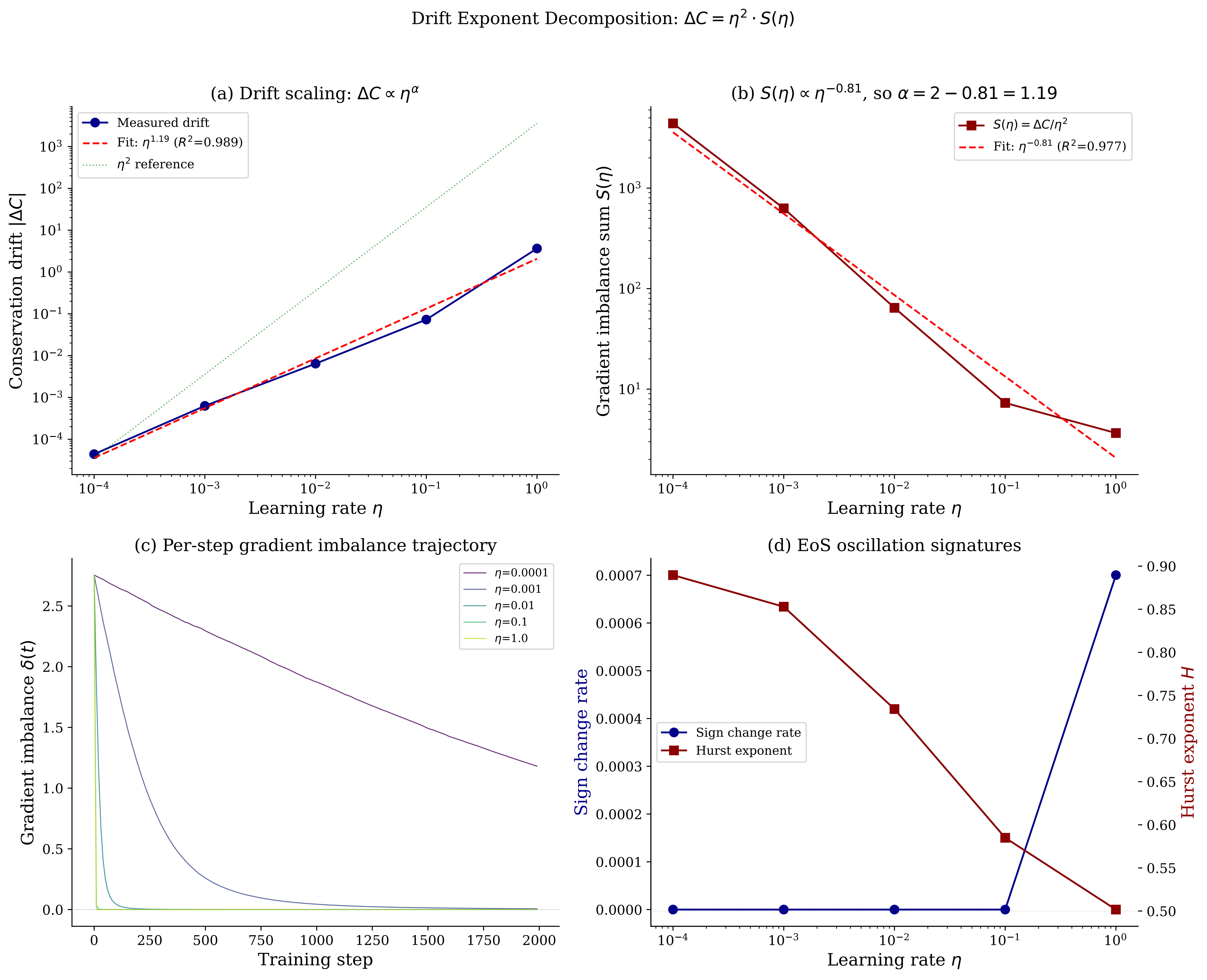

Drift Exponent Decomposition: $\Delta C = \eta^2 \cdot S(\eta)$
(a) Drift scaling: $\Delta C \propto \eta^{\alpha}$
Measured drift
Fit: $\eta^{1.19}$ (R^2=0.989)
η^2 reference
Conservation drift $|\Delta C|$
Learning rate η
(b) $S(\eta) \propto \eta^{-0.81}$, so $\alpha = 2 - 0.81 = 1.19$
$S(\eta) = \Delta C/\eta^2$
Fit: $\eta^{-0.81}$ (R^2=0.977)
Gradient imbalance sum $S(\eta)$
Learning rate η
(c) Per-step gradient imbalance trajectory
η=0.0001
η=0.001
η=0.01
η=0.1
η=1.0
Gradient imbalance $\delta(t)$
Training step
(d) EoS oscillation signatures
Sign change rate
Hurst exponent
Sign change rate
Hurst exponent H
Learning rate η